

Selecting Your First AI Use Case

Generative AI for Business Leaders & C-Suite | AI Implementation Strategy

1. Why the First Use Case Is Disproportionately Important

Your first AI use case carries weight far beyond its direct business impact. It functions as a proof point for every stakeholder watching from the sidelines. A successful first use case creates a cascade effect: skeptics become cautiously interested, fence-sitters become advocates, and leadership gains the confidence to approve larger investments. A failed first use case does the opposite with equal force. Skeptics feel vindicated, budget holders tighten purse strings, and the word 'AI' becomes associated with wasted effort for months or years.

This is not about finding the most impactful use case. It is about finding the most winnable one. The distinction matters enormously. A high-impact use case that fails teaches your organisation that AI does not work. A moderate-impact use case that succeeds teaches your organisation that AI delivers. The second lesson is infinitely more valuable at this stage, because it opens the door to everything that follows.

Think of it as landing a first client when starting a business. You do not pursue the biggest possible contract. You pursue the one you can absolutely deliver on, because that reference case opens every subsequent door.

Key Point: Your first AI use case is a political act as much as a technical one. Its primary job is not to save money or generate revenue -- it is to build organisational belief that AI works. Select for winnability above all other criteria.

2. The Three-Criteria Filter

Every potential AI use case should pass through three mandatory filters before resources are committed. These filters are designed to maximise the probability of a visible, measurable success that builds organisational momentum.

Filter 1: High Volume or Frequency

The task must be performed repeatedly -- daily or at least weekly -- by multiple people. Frequency is a multiplier on every other metric. If AI saves two hours on a task performed once a month, that is 24 hours saved annually. If AI saves two hours on a task performed daily, that is 520 hours saved annually. The same efficiency gain, multiplied 21 times by frequency alone. High-frequency tasks also provide more opportunities for learning and refinement. A team using AI on a daily task collects feedback and improves their prompts 20 times faster than a team using it monthly.

Filter 2: Clear, Quantifiable Success Metrics

You must be able to measure improvement with numbers, not feelings. This is non-negotiable because without quantifiable results, you cannot prove ROI, and without proven ROI, you cannot justify scaling.

Good Metrics (Quantifiable)	Bad Metrics (Subjective)
Time saved per task (e.g., from 4 hours to 45 minutes)	'Seems faster'
Cost per transaction reduced by X%	'Feels more efficient'
Error rate decreased from Y% to Z%	'Team likes it'
Customer satisfaction score improved by N points	'Better quality'
Response time decreased from X minutes to Y seconds	'More productive'

Filter 3: Low Risk If Imperfect

AI outputs are probabilistic, not deterministic. They are usually good but occasionally wrong. Your first use case must be one where AI mistakes have minimal consequences -- either because a human reviews the output before it reaches anyone external, or because the task itself is forgiving of errors.

Good Starting Points (Low Risk)	Bad Starting Points (High Risk)
Internal email drafts (human reviews before sending)	Legal contracts without review (errors create liability)
Meeting summaries (documentation, not decisions)	Medical diagnoses (errors endanger patients)
Content outlines and first drafts (editors refine)	Financial trading decisions (errors cost money instantly)
Data analysis with human validation	Customer-facing content without approval workflow
Internal report generation	Automated regulatory filings

3. Proven Sweet-Spot Use Cases

Research across hundreds of AI implementations reveals that approximately 70% of successful first use cases fall into five categories. These are proven starting points, not because they are the most exciting, but because they reliably pass all three filters.

Use Case Category	Why It Works	Typical Results
Email drafting and routine responses	Daily frequency, easily measured by time saved, zero risk because humans review before sending	60-80% reduction in drafting time
Meeting transcription and action items	Multiple meetings daily, measured by time saved on note-taking, zero risk (documentation only)	90% reduction in post-meeting admin
Content creation first drafts	High frequency for marketing teams, measured by output volume, low risk with editorial review	5-10x content output increase
Data analysis and insight generation	Frequent for most business teams, measured by analysis time, low risk with	70% reduction in analysis time

	human validation	
Customer support for common questions	Very high frequency, measured by response time and cost per ticket, manageable risk with escalation protocols	50-70% cost reduction per ticket

4. What to Avoid for Your First Use Case

Equally important to knowing where to start is knowing where not to start. The following categories are viable for AI but wrong for a first use case because they introduce complexity, risk, or ambiguity that undermines the primary goal of building organisational confidence.

- **Highly creative work requiring unique brand voice:** AI can assist with creative work, but replicating a distinctive brand voice or artistic vision requires significant prompt engineering and human refinement. The gap between AI output and expectation will be visible, creating a negative first impression.
- **Complex decision-making with significant consequences:** Financial, legal, or strategic decisions where errors carry real cost. The stakes are too high for a proof-of-concept, and any mistake will be used to discredit the entire AI programme.
- **Tasks requiring deep emotional intelligence:** Situations involving empathy, reading subtle human cues, or navigating sensitive interpersonal dynamics. AI has improved dramatically but is not yet reliable for emotionally complex interactions.
- **Legally regulated processes requiring explainability:** In industries where you must document how decisions were made, AI's probabilistic nature creates compliance challenges. Save these for Phase 2 when governance frameworks are established.
- **Processes under active compliance scrutiny:** If auditors are already examining a process, introducing AI adds variables that may trigger additional review cycles. Stabilise the process first, then enhance with AI.

5. The Prioritisation Matrix

Once you have identified multiple use cases that pass the three-criteria filter, the next challenge is sequencing them. The prioritisation matrix provides a visual framework for this decision.

The matrix uses two axes. The X-axis represents task frequency, from low (performed monthly or less) to high (performed daily). The Y-axis represents implementation difficulty, from easy (standard AI tools, minimal integration) to hard (custom development, system integration, change management).

Quadrant	Frequency	Difficulty	Action
Quick Wins (top priority)	High	Low	Start here immediately. These deliver fast ROI with minimal effort and build momentum.
Strategic Projects (second priority)	High	High	Tackle after quick wins build confidence. These deliver large impact but require more investment.
Fill-ins (low priority)	Low	Low	Nice to have but do not justify the

			organisational change effort in Phase 1.
Avoid in Phase 1	Low	High	High effort, low return. Do not consider until the organisation has mature AI capabilities.

Use this matrix in your Week 2 prioritisation session. Print it on a whiteboard, plot every candidate use case, and let the visual distribution guide the discussion. When the data is on the wall, the right choices become obvious and political debates give way to evidence-based decisions.

6. The Hidden Cost of Selecting the Wrong First Use Case

A poorly chosen first use case does not simply fail to deliver value. It actively damages the organisation's AI trajectory in ways that are difficult to reverse.

Real-World Incident -- IBM Watson for Oncology (2013-2022): IBM launched Watson for Oncology as a flagship AI initiative, attempting to use AI for cancer treatment recommendations -- one of the most complex, high-stakes, and emotionally charged domains imaginable. Despite billions in investment, the system struggled to match the nuanced judgment of experienced oncologists. Multiple hospitals, including MD Anderson Cancer Center, abandoned Watson after finding its recommendations were sometimes unsafe. The project became a cautionary tale that set back AI adoption in healthcare by years. The lesson for business leaders: IBM did not fail because AI cannot help with medical research. It failed because cancer treatment recommendations were the wrong first use case. Had IBM started with medical record summarisation, appointment scheduling, or insurance claim processing -- high-frequency, low-risk, clearly measurable tasks -- it would have built credibility and iteratively expanded into more complex domains.

7. The Six-Question Selection Checklist

Before committing to your first use case, answer these six questions honestly. These are binary yes/no questions. Resist the temptation to hedge with 'mostly' or 'probably'. If the answer is not a clear yes, it is a no.

Question	Why It Matters	Red Flag If 'No'
Is this task done at least weekly by multiple people?	Frequency multiplies ROI and provides rapid learning cycles.	Low-frequency tasks do not generate enough data or savings to be convincing.
Can we measure success with specific numbers?	Quantified results are the currency that buys future AI investment.	Without numbers, success becomes a matter of opinion and politics.
Are the consequences of AI errors minimal or easily caught?	First use cases must succeed. High-stakes errors can kill an entire programme.	High-consequence errors will be blamed on 'rushing into AI'.
Does the team doing this work actually want AI help?	Willing teams adopt faster, provide better feedback, and advocate to peers.	Forcing AI on a resistant team guarantees poor adoption and biased feedback.
Do we have IT and security approval for the tools	Tool approval delays can stall a pilot for weeks or months.	Start the approval process immediately or choose tools

needed?		already approved.
Can we complete a meaningful pilot in 4-6 weeks?	Short timelines force focus and prevent scope creep.	If a pilot needs longer, the use case may be too complex for Phase 1.

If you answered yes to all six, you have a winning first use case. If two or more answers are no, keep looking. Do not compromise on these criteria. A bad first use case can set your entire AI programme back by months.

8. Aligning Stakeholders on Use Case Selection

Use case selection is as much a political process as a technical one. Different stakeholders bring different perspectives, and misalignment at this stage creates friction throughout the pilot.

The CEO/Executive Sponsor

Wants visible, quotable results that justify the investment. Frame use case candidates in terms of business outcomes: 'This will reduce our proposal turnaround from two weeks to three days' rather than 'This will use GPT-4 to generate document drafts.' Executives respond to impact statements, not technology descriptions.

The IT/Security Team

Wants assurance that data is protected and systems are not compromised. Involve them early and frame their role as enabling rather than gatekeeping. Provide them with the enterprise AI tool's security documentation proactively so they can review in parallel rather than being surprised with a last-minute approval request.

The Frontline Team

Wants assurance that AI is a tool to help them, not a prelude to layoffs. Be transparent about the intention: AI handles the repetitive work they dislike so they can focus on the strategic work they find meaningful. Choose a first use case that solves a pain point the team has complained about, so AI arrives as a welcome solution rather than an imposed change.

9. From Selection to Pilot Launch

Once the use case is selected, the transition from decision to active pilot should take no more than one week. Longer transitions allow momentum to dissipate and doubts to re-emerge.

- **Day 1-2:** Appoint the pilot owner and assemble the team (3-5 people). The owner must have authority to make daily operational decisions without committee approval.
- **Day 3:** Define success metrics. Write them down, share with all stakeholders, and get explicit agreement. Ambiguous metrics at this stage become contested results later.
- **Day 4-5:** Provision AI tools. If enterprise tools are not yet approved, use the evaluation period or free tier of the tool so the team can begin immediately while formal procurement proceeds in parallel.
- **Day 6-7:** Kick-off meeting. Review the use case, tools, metrics, timeline, and escalation procedures. Start executing on Day 7. Not next week. Not after the next planning session. Day 7.

10. Key Terms Glossary

Term	Definition
Three-Criteria Filter	A selection framework requiring AI use cases to demonstrate high frequency, clear measurable metrics, and low risk of errors before being approved for piloting.
Quick Win	A use case that scores high on frequency and low on implementation difficulty, delivering fast, visible results that build organisational confidence.
Prioritisation Matrix	A 2x2 visual framework plotting use cases by frequency (X-axis) and implementation difficulty (Y-axis) to guide sequencing decisions.
Pilot Purgatory	The state where AI projects remain in indefinite testing, never progressing to scaled deployment due to lack of clear success criteria or decision-making authority.
Success Metrics	Pre-defined, quantifiable measures used to evaluate whether an AI pilot delivered its intended business value.
Probabilistic Output	The characteristic of AI systems that produce outputs based on statistical likelihood rather than deterministic rules, meaning they are usually correct but occasionally wrong.
Change Management	The structured process of preparing, equipping, and supporting people to adopt new AI-enabled ways of working.
Escalation Protocol	A defined process for routing AI-generated outputs or decisions to human reviewers when the AI encounters situations beyond its reliable capability.

11. Quick Revision Checklist

- Your first AI use case is primarily a political proof point -- select for winnability over maximum impact.
- Every use case must pass three mandatory filters: high frequency, clear quantifiable metrics, and low risk if imperfect.
- Frequency is the most powerful multiplier -- a daily task with 2 hours saved yields 520 hours annually versus 24 hours for a monthly task.
- 70% of successful first AI implementations fall into five categories: email drafting, meeting transcription, content first drafts, data analysis, and customer support.
- Avoid creative work, complex decisions, emotionally sensitive tasks, and regulated processes as first use cases.
- The prioritisation matrix plots frequency against difficulty -- always start with high-frequency, low-difficulty quick wins.
- Six yes/no questions serve as the final selection checklist -- two or more 'no' answers means keep looking.
- IBM Watson for Oncology illustrates the cost of choosing a use case that is too complex and high-stakes for initial deployment.

- Stakeholder alignment requires framing differently for executives (business outcomes), IT (security assurance), and frontline teams (pain point relief).
- Transition from selection to active pilot should take no more than one week.

20 Exam-Based MCQs — AI Use Case Selection

Test yourself after reading. These questions are written in real exam style — scenario-heavy, with plausible distractors. Read every explanation, including for questions you answered correctly.

Q1. What is the PRIMARY purpose of your first AI use case according to the lecture?

- A) Achieve the highest possible ROI to justify AI investment
- B) Build organisational belief that AI works by delivering a visible, measurable success
- C) Test the most advanced AI capabilities available
- D) Replace the most expensive employees with automation

Answer: B **Explanation:** B is correct because the lecture explicitly states that the first use case is 'a political act as much as a technical one' and its primary job is to build organisational belief. A is tempting but subordinate -- maximum ROI does not matter if the use case fails and discredits AI adoption. C prioritises technology over business outcomes. D mischaracterises AI adoption as a headcount reduction exercise.

Q2. *Scenario:* A VP of Marketing proposes using AI to completely reimagine the company's brand identity as the first AI use case. She argues that the creative potential is enormous and success would generate massive executive enthusiasm. The Head of Customer Service counter-proposes using AI to draft responses to the 200 routine customer inquiries the team handles daily. Which proposal BEST aligns with the Three-Criteria Filter?

- A) The VP's, because the creative impact would generate more enthusiasm
- B) The Head of Customer Service's, because it is high frequency (daily), clearly measurable (response time, cost per ticket), and low risk (human review before sending)
- C) Both are equally strong first use cases
- D) Neither -- the company should start with an IT infrastructure project

Answer: B **Explanation:** B is correct because customer service drafting passes all three filters: daily frequency (200 inquiries), measurable metrics (time, cost, satisfaction scores), and low risk (agents review before sending). A fails Filter 3 (brand reimagination is high-risk if AI output does not match brand standards) and Filter 2 (creative quality is hard to quantify). C ignores the clear difference in filter compliance. D introduces a category not supported by the selection framework.

Q3. Which metric is an example of a 'bad metric' for evaluating AI pilot success?

- A) Response time decreased from 11 minutes to 2 minutes
- B) Error rate reduced from 8% to 2%
- C) 'The team feels more efficient'

D) Cost per customer ticket reduced by 55%

Answer: C **Explanation:** C is correct -- 'feels more efficient' is subjective and cannot be used to prove ROI or justify scaling decisions. A, B, and D are all quantifiable metrics with specific numbers that can be validated, compared over time, and presented to budget holders.

Q4. Scenario: A pharmaceutical company is selecting its first AI use case. The options are: (1) AI-assisted drug interaction analysis for clinical trials, (2) AI-generated internal meeting summaries for research teams, and (3) AI-powered regulatory submission drafting for government agencies. Which option is the BEST first use case?

- A) Drug interaction analysis -- it has the highest potential business value
- B) Internal meeting summaries -- it is high frequency, easily measured, and carries zero risk since summaries are internal documentation
- C) Regulatory submission drafting -- it could accelerate time to market
- D) All three should be launched simultaneously as a comprehensive AI programme

Answer: B **Explanation:** B is correct because meeting summaries perfectly match all three criteria: research teams have multiple meetings daily (high frequency), time saved is easily measured (clear metrics), and internal documentation carries no regulatory or patient risk (low risk if imperfect). A has extremely high stakes -- drug interaction errors could endanger patients. C involves regulated government submissions where errors trigger compliance issues. D spreads resources too thin for a first initiative.

Q5. What is the frequency multiplier effect described in the lecture?

- A) AI tools become more accurate the more frequently they are used
- B) Tasks performed daily yield approximately 21 times more annual savings than the same task performed monthly, making frequency the most powerful ROI multiplier
- C) Teams that use AI daily are 21 times more likely to adopt it permanently
- D) High-frequency tasks require 21 times less training to automate

Answer: B **Explanation:** B is correct -- the lecture demonstrates that saving 2 hours on a daily task yields 520 hours annually versus 24 hours for a monthly task (approximately 21x). The same efficiency gain is multiplied dramatically by frequency. A describes AI learning, not the business case. C conflates adoption rate with savings. D is not supported by the content.

Q6. Scenario: A financial services firm has identified four potential first AI use cases: drafting client emails (daily, measurable, low risk), portfolio rebalancing recommendations (weekly, measurable, high risk), internal newsletter creation (monthly, hard to measure, low risk), and loan underwriting decisions (daily, measurable, high risk). Using the prioritisation matrix, which use case falls in the 'Quick Win' quadrant?

- A) Portfolio rebalancing recommendations
- B) Drafting client emails
- C) Internal newsletter creation

D) Loan underwriting decisions

Answer: B **Explanation:** B is correct -- client email drafting is high frequency (daily) and low difficulty/risk (humans review before sending), placing it squarely in the Quick Win quadrant. A is high frequency but high risk (wrong recommendations affect client portfolios). C is low frequency (monthly), placing it outside the priority quadrants. D is high frequency but high risk (underwriting errors create financial and regulatory exposure).

Q7. Why does the lecture recommend avoiding tasks requiring deep emotional intelligence as first AI use cases?

- A) AI cannot process any emotional content
- B) Emotionally sensitive interactions require reliability in reading subtle human cues, and AI is not yet consistently reliable in this domain, creating risk of highly visible failures
- C) Emotional tasks are always low frequency
- D) Employees will refuse to use AI for emotional tasks

Answer: B **Explanation:** B is correct -- the lecture identifies emotional intelligence tasks as problematic for first use cases because AI's inconsistency in reading subtle human cues creates risk of visible failures that would undermine confidence. A overstates the limitation -- AI can process emotional content, just not reliably enough for high-stakes first impressions. C is incorrect -- emotional tasks can be high frequency. D is an assumption not supported by the content.

Q8. *Scenario:* A retail company has selected 'AI-assisted inventory forecasting' as its first use case. During the pilot, the AI model produces forecasts that are 15% more accurate than the existing manual process. However, on two occasions, the AI significantly overestimated demand for seasonal products, resulting in \$200,000 in excess inventory. The CFO wants to terminate the AI programme entirely. What does this situation MOST likely indicate about the use case selection?

- A) AI is not suitable for inventory management
- B) The use case violated Filter 3 (low risk if imperfect) -- inventory forecasting errors have direct financial consequences, making it too high-stakes for a first use case
- C) The pilot was too short to produce reliable results
- D) The AI model needs more training data to eliminate errors completely

Answer: B **Explanation:** B is correct -- inventory forecasting fails Filter 3 because AI errors directly translate into excess inventory costs. Even though the AI was 15% more accurate overall, the two significant errors created a \$200,000 impact that gave the CFO ammunition to terminate the programme. A first use case where errors are visible and costly will always face this risk. A overgeneralises from one situation. C may be partially true but misses the structural issue. D assumes AI can be error-free, which contradicts the probabilistic nature of AI outputs.

Q9. According to the selection checklist, what should you do if you answer 'no' to two or more of the six selection questions?

- A) Proceed with the use case but add extra safeguards
- B) Modify the use case until all questions can be answered 'yes'
- C) Keep looking for a different use case that passes all six criteria
- D) Reduce the pilot scope to address the 'no' answers

Answer: C **Explanation:** C is correct -- the lecture states explicitly: 'If you answered no to two or more, keep looking. Do not compromise on these criteria.' The checklist is a pass/fail gate, not a negotiable guideline. A compromises on criteria the framework considers non-negotiable. B may be possible but the lecture's instruction is to keep looking rather than force-fit a case. D reduces scope but does not address fundamental filter failures.

Q10. Scenario: An insurance company's IT security team requires a 12-week evaluation before approving any new AI tool. The pilot team has identified a perfect use case (claims summary drafting) and wants to start immediately. What is the BEST approach based on the lecture's guidance?

- A) Wait the full 12 weeks before starting any pilot activities
- B) Use the AI tool's free evaluation tier to begin the pilot immediately while the formal security review proceeds in parallel
- C) Skip the security review entirely to maintain momentum
- D) Choose a different use case that does not require AI tools

Answer: B **Explanation:** B is correct -- the checklist asks 'Do we have IT and security approval for the tools needed?' and the guidance is to 'start the approval process immediately or choose tools already approved.' Using an evaluation tier while formal approval proceeds in parallel maintains momentum without bypassing security. A creates a 3-month delay that kills pilot momentum. C creates genuine security risk. D abandons a strong use case unnecessarily.

Q11. What percentage of successful first AI implementations fall into the five 'sweet spot' categories identified in the lecture?

- A) 30%
- B) 50%
- C) 70%
- D) 90%

Answer: C **Explanation:** C is correct -- the lecture states that 'these five categories account for successful first implementations in over seventy percent of companies.' A and B understate the concentration. D overstates it.

Q12. Scenario: A technology startup has selected AI-powered code review as its first use case. The engineering team is enthusiastic, but the pilot has been running for six weeks and the team cannot agree on how to measure success. Some say 'fewer bugs in production,' others say 'faster review cycles,' and the CTO says 'improved code quality' without defining what that means. What fundamental selection criterion did this pilot fail to establish?

- A) Filter 1: High frequency
- B) Filter 2: Clear, quantifiable success metrics defined before launch
- C) Filter 3: Low risk if imperfect
- D) The six-question checklist was not completed

Answer: B **Explanation:** B is correct -- the pilot failed to define clear, quantifiable success metrics before launch. The team is debating what to measure six weeks in, which means they cannot prove ROI regardless of the AI's actual performance. A is likely satisfied (code reviews happen daily). C may be satisfied depending on review workflow. D may also be true, but the specific failure described is squarely about Filter 2.

Q13. In the prioritisation matrix, which quadrant should be avoided entirely in Phase 1?

- A) High frequency, low difficulty
- B) High frequency, high difficulty
- C) Low frequency, low difficulty
- D) Low frequency, high difficulty

Answer: D **Explanation:** D is correct -- low frequency, high difficulty use cases deliver minimal return for maximum effort and should not be considered until the organisation has mature AI capabilities. A is the Quick Win quadrant (top priority). B is Strategic Projects (second priority). C is Fill-ins (low priority but not avoided).

Q14. Scenario: A consulting firm is evaluating two use cases for its first AI pilot. Use Case A: generating first drafts of client proposals (done 3 times per week, measured by drafting time, low risk with partner review). Use Case B: predicting which client accounts are at risk of churning (done monthly, measured by prediction accuracy, high consequence if wrong predictions cause inappropriate client interventions). Which use case should be selected and why?

- A) Use Case B, because client retention is more strategically important
- B) Use Case A, because it passes all three filters while Use Case B fails on frequency and risk
- C) Both should be launched as parallel pilots
- D) Neither -- consulting firms should focus on internal operations first

Answer: B **Explanation:** B is correct -- Use Case A passes all three filters (done 3x weekly, measured by time saved, partner review catches errors), while Use Case B fails Filter 1 (monthly frequency) and Filter 3 (wrong churn predictions could trigger inappropriate client interventions with real relationship consequences). A prioritises strategic importance over winnability, which the lecture explicitly warns against. C dilutes focus unnecessarily. D is not supported by the selection framework.

Q15. How does the lecture recommend framing AI use cases to executive sponsors?

- A) In terms of specific AI technology capabilities and model architectures
- B) In terms of business outcomes such as 'reduce proposal turnaround from two weeks to three days'

- C) In terms of competitive threats and market dynamics
- D) In terms of employee satisfaction improvements

Answer: B **Explanation:** B is correct -- the lecture states that executives respond to impact statements like 'reduce proposal turnaround from two weeks to three days' rather than technology descriptions. A uses technical language that does not resonate with executives. C may be relevant but is not the recommended framing approach. D is a secondary benefit, not the primary framing executives care about.

Q16. Scenario: A healthcare organisation wants to use AI to generate patient discharge summaries. The task is performed 50 times daily, time savings are easily measured, and nurses review every summary before it reaches the patient. However, the nursing staff has expressed strong resistance, stating they believe AI will eventually replace their jobs. Which selection checklist question is at risk of failing?

- A) 'Is this task done at least weekly by multiple people?'
- B) 'Can we measure success with specific numbers?'
- C) 'Does the team doing this work actually want AI help?'
- D) 'Can we complete a meaningful pilot in 4-6 weeks?'

Answer: C **Explanation:** C is correct -- the nursing staff has expressed strong resistance based on job security fears. The lecture warns that 'forcing AI on a resistant team guarantees poor adoption and biased feedback.' A is clearly passed (50 times daily). B is clearly passed (time savings are measurable). D is likely feasible. The team willingness question is the critical risk factor here.

Q17. What lesson does the IBM Watson for Oncology case illustrate about first use case selection?

- A) AI cannot be used in healthcare under any circumstances
- B) Large companies have unique challenges that make AI adoption harder
- C) Choosing a highly complex, high-stakes domain as a first use case creates risk of visible failure that damages AI credibility for years
- D) IBM's technology was not advanced enough for medical applications

Answer: C **Explanation:** C is correct -- IBM Watson for Oncology illustrates that even with billions in investment, choosing cancer treatment recommendations (extremely complex, extremely high stakes) as a first use case created public failures that damaged AI credibility across healthcare. A overgeneralises -- AI is being used successfully in many healthcare applications. B misattributes the problem to company size rather than use case selection. D oversimplifies; the technology was capable but the application was wrong for a first proof point.

Q18. Scenario: After selecting a first AI use case (internal email drafting), a company's pilot owner wants to expand the scope to include external client communications, proposal writing, and contract amendments -- all within the first pilot. What is the PRIMARY risk of this scope expansion?

- A) The expanded scope introduces high-risk tasks (client communications, contracts) that violate Filter 3, converting a low-risk quick win into a high-stakes experiment
- B) The team does not have enough people to handle the expanded scope

- C) AI tools cannot handle multiple use cases simultaneously
- D) The pilot timeline will need to be extended to 12 weeks

Answer: A **Explanation:** A is correct -- the original use case (internal email drafting) is low risk because errors stay internal. Adding external client communications and contract amendments introduces situations where AI errors have real business consequences, fundamentally changing the risk profile. B may be true but is secondary. C is incorrect -- AI tools can handle multiple tasks. D is a consequence, not the primary risk.

Q19. According to the lecture, how long should the transition from use case selection to active pilot execution take?

- A) One day
- B) One week
- C) One month
- D) One quarter

Answer: B **Explanation:** B is correct -- the lecture outlines a 7-day transition from selection to execution: Days 1-2 for team formation, Day 3 for metrics definition, Days 4-5 for tool provisioning, and Days 6-7 for kick-off and first execution. A is unrealistically fast. C and D allow momentum to dissipate and doubts to re-emerge.

Q20. Scenario: A manufacturing company has narrowed its first AI use case candidates to two options. Option 1: AI-generated quality inspection reports (done hourly by 15 technicians, currently takes 20 minutes per report, easily measured). Option 2: AI-predicted equipment maintenance schedules (done weekly, requires integration with IoT sensor data, higher accuracy could prevent \$500K in downtime). Using the prioritisation matrix, which option should be selected FIRST?

- A) Option 2, because the financial impact of preventing downtime is much larger
- B) Option 1, because it is high frequency and low difficulty (Quick Win quadrant), while Option 2 is lower frequency and higher difficulty (Strategic Project quadrant)
- C) Both should be pursued simultaneously
- D) Neither -- manufacturing should focus on supply chain AI applications first

Answer: B **Explanation:** B is correct -- Option 1 is performed hourly by 15 technicians (extremely high frequency), uses standard AI text generation (low difficulty), and produces internal reports (low risk). It sits squarely in the Quick Win quadrant. Option 2 has larger potential impact but is lower frequency (weekly), requires IoT integration (higher difficulty), and has significant consequences if predictions are wrong (higher risk). The prioritisation matrix says to start with quick wins. A prioritises potential impact over winnability, which the lecture warns against. C spreads resources unnecessarily. D is not supported by the framework.